SODIUM VALPROATE

Highly teratogenic — Pregnancy Prevention Programme

Hepatotoxic & pancreatitis

CHEMICAL NAME Sodium 2-propylpentanoate (valproic acid sodium salt)
CLASS Broad-spectrum antiepileptic and mood stabiliser
TRADE NAMES Epilim · Depakine · Depakote (divalproex) · Convulex

THERAPEUTIC USES / INDICATIONS

- Generalised epilepsies: tonic-clonic, absence, myoclonic (drug of choice in men).
- Focal seizures; status epilepticus (IV, second-line).
- Bipolar disorder (acute mania, prophylaxis).
- Migraine prophylaxis (off-label in some countries).

ADMINISTRATION

- **PO** with or after food; enteric-coated/MR tablets swallowed whole.
- **IV** injection over 3–5 min or infusion.
- Women/girls of childbearing potential: only if other treatments unsuitable, under Pregnancy Prevention Programme (annual risk form)

DOSAGES

- **Epilepsy (adult):** start 600 mg/day in divided doses; increase by 200 mg every 3 days; usual 1000–2000 mg/day (20–30 mg/kg/day), max 2500 mg.
- **Status epilepticus:** 40 mg/kg IV (max 3000 mg) over 10 min (ESETT).
- **Children:** 10–20 mg/kg/day, maintenance up to 40 mg/kg/day.
- **Mania (valproate semisodium):** 750 mg/day in divided doses, increased quickly to response (max 60 mg/kg/day).

PHARMACOKINETICS

- **Peak:** 1–4 h (EC delayed). Highly protein-bound (saturable).
- Metabolised in liver (glucuronidation, β-oxidation); enzyme inhibitor.
- Excreted in urine. **t½:** 8–20 h. Levels (50–100 mg/L) poorly correlate with effect.

PHARMACODYNAMICS

- ↑ brain **GABA** (inhibits GABA transaminase), blocks **sodium channels** and T-type **calcium channels**.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, dyspepsia.
- Weight gain, hair loss (regrows curly).
- Tremor, drowsiness.

ADVERSE EFFECTS (SERIOUS)

- Teratogenicity: neural tube defects (1–2%), major malformations (~10%), neurodevelopmental disorders/autism (30–40%).
- Hepatotoxicity (fatal, especially <3 yr, metabolic disorders, first 6 months).
- Pancreatitis; hyperammonaemic encephalopathy.
- Thrombocytopenia, platelet dysfunction; PCOS; Fanconi syndrome.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (epilepsy) unless no alternative; women of childbearing potential without PPP; pregnancy (bipolar, migraine) — contraindicated.
- Active liver disease; family history of severe hepatic dysfunction.
- Mitochondrial disorders (POLG); urea cycle disorders; porphyria.

KEY INTERACTIONS

- Lamotrigine: ↑ lamotrigine levels (halve its dose); SJS risk.
- Carbapenems (meropenem): ↓↓ valproate levels → seizures.
- **Phenobarbital, phenytoin:** ↑ their levels; **carbamazepine:** ↑ epoxide.
- **Aspirin:** ↑ free valproate; **topiramate:** hyperammonaemia.

NURSING MANAGEMENT

- Confirm pregnancy prevention programme for females; pregnancy test.
- LFTs and CBC (platelets) before starting, at intervals in first 6 months.
- Report vomiting, abdominal pain, lethargy, jaundice (hepatotoxicity/pancreatitis).
- Check ammonia if confusion.
- Check for carbapenem prescription.

PATIENT EDUCATION

- Women: use highly effective contraception; never stop suddenly if pregnant — see doctor urgently.
- Report abdominal pain, vomiting, yellow skin, unusual bruising or drowsiness.
- Take with food; weight management advice.
- Do not stop suddenly.

KEY POINTS

- Broad-spectrum antiepileptic (GABA ↑).
- Hepatotoxicity, pancreatitis, hyperammonaemia.
- Highly teratogenic; PPP for women.
- Raises lamotrigine; destroyed by carbapenems.

SOURCES CHECKED **[724]** Epilim 200 mg Gastro-resistant tablets (UK SmPC) **[725]** Kapur et al., N Engl J Med 2019 **[726]** valproic acid, oral (US label) **[727]** Depakote, oral (US label) **[728]** Valproic Acid, oral (US label)

Full references p. 432–457